Chapter 4, Section 4

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3. Let the elliptic curve X be embedded in $\mathbb{P}^2$ so as to have the equation $y^2 = x(x-1)(x-\lambda)$. Show that any automorphism of X leaving $P_0 = (0:1:0)$ fixed is induced by an automorphism of $\mathbb{P}^2$ coming from the automorphism of the affine (x, y)-plane given by

$$\begin{cases} x' = ax + b \\ y' = cy. \end{cases}$$

In each of the four cases of (4.7), describe these automorphisms of $\mathbb{P}^2$ explicitly, and hence determine the structure of the group $G = \mathrm{Aut}(X, P_0)$.

Proof. Let $: X \to \mathbb{P}^1$ be the unique g^1_2 with $f(P_0) = \infty$, branched over $0, 1, \lambda, \infty$. If $\sigma \in G$, then by (4.4) there exists an automorphism τ of $\mathbb{P}^1$ which fixes ∞ and satisfies $f \circ \sigma = \tau \circ f$. In particular, τ sends $\{0, 1, \lambda\}$ to $\{0, 1, \lambda\}$ in some order. If $\tau = \mathrm{id}$, then either $\sigma = \mathrm{id}$ or σ is the unique automorphism of order 2 fixing P_0 and interchanging the sheets of f. We call it the *hyperbolic involution* of X, and we denote it by $\iota : X \to X$. It comes from the automorphism of the affine (x, y)-plane corresponding to the reflection across the x-axis:

$$\begin{cases} x' = x \\ y' = -y \end{cases}.$$

The other automorphims are determined using the methods of (4.7). In particular, we look at the orbits of the action of S_3 on $k - \{0, 1\}$ (4.5).

$\boxed{j \neq 0, 1728}$ Since the orbits of λ are all distinct, $\tau = \mathrm{id}$, and $\sigma = \mathrm{id}$ or ι.

$\boxed{j = 1728, \mathrm{char}\, k \neq 3}$ This is the case $\lambda = -1, \frac{1}{2}$, or 2, and λ coincides with one other element of its orbit under the action of S_3. For example, if $\lambda = -1$, then $\lambda = 1/\lambda$, and G is the cyclic group of order 4 generated by

$$\begin{cases} x' = -x \\ y' = iy \end{cases}.$$

$\boxed{j = 0, \mathrm{char}\, k \neq 3}$ This is the case $\lambda = -\omega$ or $-\omega^2$, where ω is a third root of unity, and λ coincides with two other elements of its orbit under S_3. For example, if $\lambda = -\omega$, then

$$\lambda = \frac{1}{1-\lambda} = \frac{\lambda - 1}{\lambda},$$

and G is the cyclic group of order 6 generated by

$$\begin{cases} x' = \omega^2 x + 1 \\ y' = -y \end{cases}.$$

$\boxed{j = 0\ (= 1728), \mathrm{char}\, k = 3}$ If $\mathrm{char}\, k = 3$ and $j = 0$, then $\lambda = -1$, and all six elements of its orbit under S_3 coincide. Note that k has no nontrivial third roots of unity, and the values $\lambda = -1, \frac{1}{2}, 2$ of the previous case coincide too. So G is the cyclic group of order 12 generated by

$$\begin{cases} x' = x + 1 \\ y' = iy \end{cases}.$$

$\square$

4. Let X be an elliptic curve in $\mathbb{P}^2$ given by an equation of the form

$$y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6.$$

Show that the j-invariant is a rational function of the a_i, with coefficients in $\mathbb{Q}$. In particular, if the a_i are all in some field $k_0 \subseteq k$, then $j \in k_0$ also. Furthermore, for every $\alpha \in k_0$, there exists an elliptic curve defined over k_0 with j-invariant equal to α.

Proof. Complete the square in y to get an equation of the form

$$y^2 = x^3 + Ax^2 + Bx + C,$$

with

$$A = \frac{a_1^2}{4} + a_2, \quad B = \frac{a_1a_3}{2} + a_4, \quad C = \frac{a_3^2}{4} + a_6.$$

Let α, β, γ be the distinct roots of the cubic on the right-hand side. Applying a linear transformation of x, which sends α, β to $0, 1$, transforms the equation of X into

$$y^2 = x(x-1)(x-\lambda), \quad \lambda = \frac{\gamma - \alpha}{\beta - \alpha}.$$

We can plug in λ into the j-invariant formula. Simplifying, we get a rational function of the elementary symmetric polynomials of α, β, γ (which are just A, B, C) with coefficients in $\mathbb{Q}$

$$j = 2^8 \frac{(A^2 - 3B)^3}{A^2B^2 - 4B^3 - 4A^3 + 18ABC - 27C^2}.$$

Since A, B, C are polynomials of the a_i with integer coefficients, the j-invariant is a rational function of the a_i with coefficients in $\mathbb{Q}$. □

5. Let X, P_0 be an elliptic curve having an endormorphism $f : X \to X$ of degree 2.

(a) If we represent X as a 2-1 covering of $\mathbb{P}^1$ by a morphism $\pi : X \to \mathbb{P}^1$ ramified at P_0, then as in (4.4), show that there is another morphism $\pi' : X \to \mathbb{P}^1$ and a morphism $g : \mathbb{P}^1 \to \mathbb{P}^1$, also of degree 2, such that $\pi \circ f = g \circ \pi'$.

(b) For a suitable choice of coordinates in the two copies of $\mathbb{P}^1$, show that g can be taken to be the morphism $x \mapsto x^2$.

(c) Now show that g is branched over two of the branch points of π, and that g^{-1} of the other two branch points of π consists of the four branch points of π'. Deduce a relation involving the invariant λ of X.

(d) Solving the above, show that there are just three values of j corresponding to elliptic curves with an endomorphism of degree 2, and find the corresponding values of λ and j.

Proof.

(a) The morphism $\pi \circ f$ corresponds to the linear system $|f^*(2P_0)|$. Notice that we can write $f^*(2P_0) = 2f^*(P_0)$, which gives a hint as to how to factor $\pi \circ f$. In particular, we can factor $\pi \circ f$ through the morphism $\pi' : X \to \mathbb{P}^1$ given by the linear system $|f^*(P_0)|$ on X, followed by the morphism $g : \mathbb{P}^1 \to \mathbb{P}^1$ given by the linear system $|\pi'_*(f^*P_0)|$ on $\mathbb{P}^1$.

(b) Any degree two morphism $g : \mathbb{P}^1 \to \mathbb{P}^1$ is branched over two points, and we can choose coordinates in the two copies of $\mathbb{P}^1$ so that these two points are $0, \infty$ in the target and source. Then g is given by the morphism $x \mapsto x^2$. Indeed, on some local affine patch, g is given by a degree two polynomial in x, say $g(x) = (x - \alpha)(x - \beta)$. Then g is ramified at $x = \frac{1}{2}(\alpha + \beta)$ and ∞, and we can choose coordinates so that these two points are $0, \infty$. Similarly, on the target, we can choose coordinates so that the two branch points of g are $0, \infty$. Then g is given by a degree two polynomial with roots at $0, \infty$, which is just $x \mapsto x^2$.

(c) The morphism $\pi \circ f$ is generically four-to-one, and it is branched over the branch points of π because any morphism of elliptic curves is unramified by the Riemann-Hurwitz formula. This just says $(\pi \circ f)^{-1}$ of a branch point of π consists of two points. Since g is a morphism of degree two ramified at exactly two points, and π' is ramified at exactly four points, g must be branched over two of the branch points of π, and g^{-1} of the other two branch points of π consists of the four branch points of π'. Note that one of the points g is branched over is given by

$$g(\pi'(f^{-1}P_0)) = \pi(f(f^{-1}P_0)) = \pi(P_0).$$

Now, choose coordinates in the two copies of $\mathbb{P}^1$ so that the branch points of g are $0, \infty$, the other two branch points of π are $1, \lambda$, and g is given by $x \mapsto x^2$. Then the four branch points of π' are $\pm 1, \pm\sqrt{\lambda}$. Consider the linear transformation of $\mathbb{P}^1$ sending $-1, 1, \sqrt{\lambda}$ to $0, 1, \infty$, respectively, which is given by

$$\phi(x) = \frac{(1-\sqrt{\lambda})(x+1)}{2(x-\sqrt{\lambda})}$$

Then the branch points of π' are sent to $0, 1, \infty, \phi(-\sqrt{\lambda})$, where

$$\lambda' := \phi(-\sqrt{\lambda}) = -\frac{(1-\sqrt{\lambda})^2}{4\sqrt{\lambda}}.$$

Since λ' and λ give the same j-invariant, they must be related by a fractional linear transformation in S_3. In particular, λ' is in the orbit of λ under the action of S_3 on $k - \{0, 1\}$ (4.5).

(d) Solving for λ in the above, we get the following three unique values of λ modulo the action of S_3:

$$\lambda = 2, \quad 3 + 2\sqrt{2}, \quad \frac{1 + 3i\sqrt{7}}{2}.$$

The corresponding values of j are

$$j = 2^6 \cdot 3^3, \quad 2^6 \cdot 5^3, \quad -3^3 \cdot 5^3.$$

□

7. *The Dual of a Morphism.* Let X and X' be elliptic curves over k, with base points P_0, P_0'.

(a) If $f : X \to X'$ is any morphism, use (4.11) to show that $f^* : \operatorname{Pic} X' \to \operatorname{Pic} X$ induces a homomorphism $\hat{f} : (X', P_0') \to (X, P_0)$. We call this the *dual* of f.

(b) If $f : X \to X'$ and $g : X' \to X''$ are two morphisms, then $\widehat{g \circ f} = \hat{f} \circ \hat{g}$.

(c) Assume $f(P_0) = P_0'$, and let $n = \deg f$. Show that if $Q \in X$ is any point, and $f(Q) = Q'$, then $\hat{f}(Q') = n_X(Q)$. Conclude that $f \circ \hat{f} = n_{X'}$ and $\hat{f} \circ f = n_X$.

(d) If $f, g : X \to X'$ are two morphisms preserving the base points P_0, P_0', then $\widehat{f+g} = \hat{f} + \hat{g}$.

(e) Using (d), show that for any $n \in \mathbb{Z}$, $\hat{n}_X = n_X$. Conclude that $\deg n_X = n^2$.

(f) Show for any f that $\deg \hat{f} = \deg f$.

Proof. Let the pair $(X, \mathcal{P})$ (resp. $(X', \mathcal{P}')$) be the Jacobian of (X, P_0) (resp. (X', P_0')). If $\mathcal{C}$ is any category, and T is any object in $\mathcal{C}$, then denote by $h_T : \mathcal{C}^{op} \to \mathbf{Set}$ the contravariant Hom functor defined by $U \mapsto \operatorname{Hom}_{\mathcal{C}}(U, T)$. Recall that X and $\mathcal{P} \in \operatorname{Pic}^\circ(X/X)$ satisfies the following universal property: if T is any scheme of finite type over k, and for any $\mathcal{M} \in \operatorname{Pic}^\circ(X/T)$, there is a unique morphism $f : T \to X$ such that $f^*\mathcal{P} \cong \mathcal{M}$ in $\operatorname{Pic}^\circ(X/T)$. In other words, $(X, \mathcal{P})$ represents the functor $\operatorname{Pic}^\circ(X/-)$. Lastly, since $(X, \mathcal{P})$ is a group object in the category of schemes of finite type over k, the set $h_X(T)$ has a group structure induced by $\operatorname{Pic}^\circ(X/T)$.

(a) Any morphism $f : X \to X'$ defines a natural transformation $f^* : \operatorname{Pic}^\circ(X'/-) \to \operatorname{Pic}^\circ(X/-)$ of group functors in the following way (4.10.2).

If T is any scheme of finite type over k, then $f' = f \times 1_T : X' \times T \to X \times T$ induces a group homomorphism $f^* : \operatorname{Pic}(X' \times T) \to \operatorname{Pic}(X \times T)$. Let $p : X \times T \to T$ and $p' : X' \times T \to T$ be the projection maps. The image of f^* lies in $\operatorname{Pic}^\circ(X \times T)$, and $p = p' \circ f'$ implies $f^* p'^* \operatorname{Pic}(T) = p^* \operatorname{Pic}(T)$. Hence, we have a well-defined group homomomorphism $f_T^* : \operatorname{Pic}^\circ(X'/T) \to \operatorname{Pic}^\circ(X/T)$ for each scheme T of finite type over k.

It remains to show f^* is functorial. Let $g : S \to T$ be any morphism of schemes of finite type over k. We can apply $\operatorname{Pic}(-)$ to the following commutative diagram

$$\begin{array}{ccc} X \times S & \xrightarrow{f \times 1_S} & X' \times S \\ {\scriptstyle 1_X \times g}\downarrow & & \downarrow{\scriptstyle 1_{X'} \times g} \\ X \times T & \xrightarrow{f \times 1_T} & X' \times T \end{array},$$

which gives a commutative diagram of Picard groups with arrows given by taking inverse images of invertible sheaves.

Now, we identify h_X and $h_{X'}$ with $\mathrm{Pic}^\circ(X/-)$ and $\mathrm{Pic}^\circ(X'/-)$, respectively. Yoneda's lemma says there exists a natural bijection between the set of morphisms $X' \to X$ and the set of natural transformations of functors $h_{X'} \to h_X$. Hence, there exists a unique morphism $\hat{f} \in \mathrm{Hom}(X', X)$ such that $f^* \in \mathrm{Hom}(h_{X'}, h_X)$ is given by composition with $\hat{f}$. The invertible sheaves $\mathcal{O}_X$ and $\mathcal{O}_{X'}$ correspond to $P_0 \in X$ and $P_0' \in X'$, respectively, and structure sheaves pullback, so we conclude that $\hat{f}(P_0') = P_0$.

(b) This is a consequence of the uniqueness property of $\hat{f}$ in $\mathrm{Hom}(X', X)$, and the analogous fact for inverse image of sheaves, i.e., $(g \circ f)^* = f^* \circ g^*$.

(c) The points $Q' \in X'$ and $\hat{f}(Q') \in X$ correspond to $\mathcal{O}_{X'}(Q' - P_0')$ in $\mathrm{Pic}^\circ(X'/k)$ and $f^*\mathcal{O}_{X'}(Q' - P_0')$ in $\mathrm{Pic}^\circ(X/k)$, respectively. Thus, it is enough to show the linear equivalence of the following divisors

$$f^*(Q' - P_0') \sim n_X(Q) - P_0.$$

By the hypothesis, $Q' - P_0' = f_*(Q - P_0)$, which by (Ex. 2.6) implies

$$f^*(Q' - P_0') = n(Q - P_0).$$

The right-hand side is linearly equivalent to $n_X(Q) - P_0$ because the group law on X is induced by that of $\mathrm{Pic}^\circ(X/k)$, so we conclude that $\hat{f} \circ f = n_X$. For the second corollary, observe that f and multiplication by n commutes since f is a morphism of proper group schemes. Hence, we have

$$(f \circ \hat{f})(Q') = f(n_X(Q)) = n_{X'}(f(Q)) = n_{X'}(Q').$$

(d) It is enough to show for any $\mathcal{L}' \in \mathrm{Pic}^\circ(X')$, that

$$(f + g)^*\mathcal{L}' \cong f^*\mathcal{L}' \otimes g^*\mathcal{L}'.$$

Let Q' be the closed point in X' such that $\mathcal{L}' \cong \mathcal{O}_{X'}(Q' - P_0')$. There exists a bijection

$$\begin{aligned} \mathrm{Hom}_{\mathbf{Sch}(k)}(X, X') &\xrightarrow{\sim} \mathrm{Pic}^\circ(X'/X) \\ f : X \to X' &\mapsto \mathcal{L}_f := (f \times 1_{X'})^*\mathcal{P}', \end{aligned}$$

where the group law on the left-hand side is induced by $\mathrm{Pic}^\circ(X'/X)$, i.e., $\mathcal{L}_{f+g} = \mathcal{L}_f \otimes \mathcal{L}_g$ is a representative of $f + g$ in $\mathrm{Pic}(X \times X')$ for any $f, g : X \to X'$. Let $\sigma : X \to X \times X'$ be the section $x \mapsto (x, Q')$. If we assume $\mathcal{P}' = \mathcal{O}_{X' \times X'}(\Delta - X' \times P_0')$ as in (4.11), then $\sigma^*\mathcal{L}_f \cong f^*\mathcal{O}_{X'}(Q')$, so replacing $\mathcal{P}'$ by $\mathcal{P}' \otimes p_2^*\mathcal{O}_{X'}(-P_0)$, we get $\sigma^*\mathcal{L}_f \cong f^*\mathcal{L}'$. Hence,

$$(f + g)^*\mathcal{L}' \cong \sigma^*\mathcal{L}_{f+g} \cong \sigma^*\mathcal{L}_f \otimes \sigma^*\mathcal{L}_g \cong f^*\mathcal{L}' \otimes g^*\mathcal{L}'.$$

(e) Proceeding by induction on n reduces the statement to the case $n = 2$. The dual of the identity is the identity. Hence, $\hat{2}_X = \widehat{1_X + 1_X} = 1_X + 1_X = 2_X$ by (d). According to (c), $\hat{n}_X \circ n_X$ is multiplication by $\deg n_X$. Since $\hat{n}_X \circ n_X = n_X^2$, we conclude that $\deg n_X = n^2$.

(f) By (c) and (e), $\deg \hat{f} \circ f = n^2$. The degree of morphisms of curves is multiplicative with respect to composition. Hence, $\deg \hat{f} = n$.

□

9. We say two elliptic curves X, X' are *isogenous* if there is a finite morphism $f : X \to X'$.

(a) Show that isogeny is an equivalence relation.

(b) For any elliptic curve X, show that the set of elliptic curves X' isogenous to X, up to isomorphism is countable.

Proof.

(a) The identity morphism is trivially finite, so any elliptic curve is isogenous to itself. A composition of finite morphisms is finite, so isogeny satisfies transitivty. Lastly, (Ex. 4.7) implies isogeny is reflexive. Hence, isogeny is an equivalence relation.

(b) Since $\ker f$ is a closed subgroup scheme of X, it is a finite set of points, all of which are torsion under the group law of X. Every torsion point of X lies in the kernel of a multiplication map $n_X : X \to X$ for some $n > 0$. By (Ex. 4.7), n_X is an étale covering of degree n^2, which implies X has exactly n^2 points of order n. Hence, there are countably many torsion points on X. The set of finite subsets of a countable set is countable, so there are only countably many candidates for $\ker f$. Since X' is uniquely determined by X and $\ker f$, the set of elliptic curves isogenous to X is countable.

□

11. Let X be an elliptic curve over $\mathbb{C}$, defined by the elliptic functions with periods $1, \tau$. Let R be the the ring of endormorphisms of X.

(a) If $f \in R$ is a nonzero endormorphism correspondending to complex multiplication by α as in (4.18), show that $\deg f = |\alpha|^2$.

(b) If $f \in R$ corresponds to $\alpha \in \mathbb{C}$ again, show that the dual $\hat{f}$ of (Ex. 4.7) corresponds to the complex conjugate $\bar{\alpha}$ of α.

(c) If $\tau \in \mathbb{Q}(\sqrt{-d})$ happens to be integral over $\mathbb{Z}$, show that $R = \mathbb{Z}[\tau]$.

Proof.

(a) It is not hard to see $\deg f$ is the size of the quotient $|\Lambda/\alpha\Lambda|$. Equivalently, it is the ratio between the areas of the parallelograms with vertices $\alpha, \alpha\tau$, and $1, \tau$. But α just scales the area by $|\alpha|^2$. Hence, $\deg f = |\alpha|^2$.

(b) By (Ex. 7c), $\hat{f} \circ f$ is multiplication by $|\alpha|^2$, but f is multiplication by α, and $\bar{\alpha}$ is the unique element in $\mathbb{C}$ such that $\alpha\bar{\alpha} = |\alpha|^2$. Hence, $\hat{f}$ is multiplication by $\bar{\alpha}$.

(c) The ring of integers of $\mathbb{Q}(\sqrt{-d})$ is given by

$$\begin{cases} \mathbb{Z}[\sqrt{-d}] & d \equiv 1, 2 \mod 4 \\ \mathbb{Z}\left[\dfrac{1+\sqrt{-d}}{2}\right] & d \equiv 3 \mod 4 \end{cases}.$$

By (4.19), if $\tau = r + s\sqrt{-d}$ for some $r, s \in \mathbb{Q}$, the endomorphism ring R is

$$R = \{a + b\tau \mid a, b \in \mathbb{Z}, \quad 2br, b(r^2 + ds^2) \in \mathbb{Z}\}.$$

Suppose τ is integral over $\mathbb{Z}$. If $d \equiv 1, 2 \mod 4$, then $r, s \in \mathbb{Z}$, so $a + b\tau \in R$ for all $a, b \in \mathbb{Z}$. If $d \equiv 3 \mod 4$, then we can write $\tau = (r' + s'\sqrt{-d})/2$ with $r', s' \in \mathbb{Z}$ such that $r' \equiv s' \mod 2$. For any $b \in \mathbb{Z}$, the first condition $2br = br' \in \mathbb{Z}$ is easily satisfied. Also, we have

$$r^2 + ds^2 = \frac{r'^2 + ds'^2}{4} \in \mathbb{Z}$$

for all $b \in \mathbb{Z}$ since $r'^2, s'^2 \equiv 0, 1 \mod 4$. Hence, $R = \mathbb{Z}[\tau]$.

□

12. Again let X be an elliptic curve over $\mathbb{C}$ determined by the elliptic functions with periods $1, \tau$, and assume that τ lies in the region G of (4.15B).

(a) If X has any autmorphisms leaving P_0 fixed other than ± 1, show that either $\tau = i$ or $\tau = \omega$, as in (4.20.1) and (4.20.2). This gives another proof of the fact (4.7) that there are only two curves, up to isomorphism, having automorphisms other than ± 1.

(b) Now show that there are exactly three values of τ for which X admits an endomorphism of degree 2. Can you match these with the three values of j determined in (Ex. 4.5)?

Proof.

(a) By (Ex. 4.11), τ must be a root of unity in G. The only roots of unity in G are i and ω.

(b) By (Ex. 4.11), we need to find the values of τ such that Λ contains an element of norm 2. In particular, if we write $m + n\tau \in \Lambda$, then we need to determine when

$$m^2 + n^2|\tau|^2 + 2mn\operatorname{Re}(\tau) = 4$$

admits a solution for some $m, n \in \mathbb{Z}$. Using the bounds on $|\tau|$ and $\operatorname{Re}(\tau)$ as specified in (4.15B), the only possible values of τ are

$$\tau = i, \quad i\sqrt{2}, \quad -\frac{1}{2} + i\frac{\sqrt{7}}{2}.$$

The corresponding values of j are

$$j = 2^6 \cdot 3^3, \quad 2^6 \cdot 5^3, \quad -3^3 \cdot 5^3.$$

□

13. If $p = 13$, there is just one value of j for which the Hasse invariant of the corresponding curve is 0. Find it.

Proof. The polynomial $h_p(\lambda)$ of (4.22) for $p = 13$ is

$$h_{13}(\lambda) = \lambda^6 + 10\lambda^5 + 4\lambda^4 + 10\lambda^3 + 4\lambda^2 + 10\lambda + 1,$$

and an elliptic curve has Hasse invariant 0 if and only if $h_{13}(\lambda) = 0$. On the otherhand, consider the j-invariant

$$j = 2^8 \frac{(\lambda^2 - \lambda + 1)^3}{\lambda^2(\lambda - 1)^2}.$$

Expanding above modulo 13 gives

$$\lambda^6 + 10\lambda^5 + (10j + 6)\lambda^4 + (6j + 6)\lambda^3 + (10j + 6)\lambda^2 + 10\lambda + 1 = 0.$$

Matching coefficients with $h_{13}(\lambda)$ and solving for j gives $j \equiv 5 \mod 13$. □

14. The Fermat curve $X : x^3 + y^3 = z^3$ gives a nonsingular curve in characteristic p for every $p \neq 3$. Determine the set $\mathfrak{P} = \{p \neq 3 \mid X_{(p)}$ has Hasse invariant $0\}$, and observe (modulo Dirichlet's theorem) that it is a set of primes of density $1/2$.

Proof. Using (4.21) with $f(x, y, z) = x^3 + y^3 - z^3$, we find that the coefficient of $(xyz)^{p-1}$ in f^{p-1} is zero if and only if $p \equiv 2 \mod 3$. By Dirichlet's theorem, the density of $\mathfrak{P}$ is $1/\phi(3) = 1/2$. □

15. Let X be an elliptic curve over a field k of characteristic p. Let $F' : X_p \to X$ be the k-linear Frobenius morphism (2.4.1). Use (4.10.7) to show that the dual morphism $\hat{F}' : X \to X_p$ is separable if and only if the Hasse invariant of X is 1. Now use (Ex. 4.7) to show that if the Hasse invariant is 1, then the subgroup of points of order p on X is isomorphic to $\mathbb{Z}/p$; if the Hasse invariant is 0, then it is 0.

Proof. By (III, Ex. 8.1) and (4.10.7), if $f : X \to X'$ is any morphism of elliptic curves over k, then the induced map of cohomology $f^* : H^1(X, \mathcal{O}_{X'}) \to H^1(X, \mathcal{O}_X)$ is precisely the map of tangent spaces at 0 of the dual map $\hat{f} : X' \to X$. By the Riemann-Hurwitz formula, any separable morphism of elliptic curves is not ramified. Thus, a morphism of elliptic curves $f : X \to X'$ is inseparable if and only if f is ramified at a point, in which case f is ramified everywhere by the group structure of elliptic curves. Also, any morphism of curves is ramified at a point if and only if the map of tangent spaces is not an isomorphism. Hence, $\hat{F}'$ is separable if and only if the map of tangent spaces $F'^* : H^1(X, \mathcal{O}_X) \to H^1(X_p, \mathcal{O}_{X_p})$ is not the zero map if and only if the Hasse invariant of X is 1.

The subgroup of points of order p on X is the kernel of the multiplication by p morphism $p_X : X \to X$. Since the k-linear Frobenius morphism F' has degree p, we have $p_X = F' \circ \hat{F}'$ by (Ex. 4.7c). Also, F' is a homeomorphism on the underlying topological spaces, i.e., F' is a one-to-one map. If the Hasse invariant of X is 1, then $\hat{F}'$ is a separable morphism of degree p, so $\hat{F}'$ is a p-to-one map. Hence, p_X is a p-to-one map, and $(\ker p_X)[k] \cong \mathbb{Z}/p$. If the Hasse invariant of X is 0, then $p_X = F' \circ \hat{F}'$ is a purely inseparable morphism of curves. Hence, p_X is a one-to-one map, and $(\ker p_X)[k] = 0$. □

16. Again let X be an elliptic curve over k of characteristic p, and suppose X is defined over the field $\mathbb{F}_q$ of $q = p^r$ elements, i.e., $X \subseteq \mathbb{P}^2$ can be defined by an equation with coefficients in $\mathbb{F}_q$. Assume also that X has a rational point over $\mathbb{F}_q$. Let $F' : X_q \to X$ be the k-linear Frobenius with respect to q.

(a) Show that $X_q \cong X$ as schemes over k, and that under this identification, $F' : X_q \to X$ is the map obtained by the qth-power map on the coordinates of points of X, embedded in $\mathbb{P}^2$.

(b) Show that $1_X - F'$ is a separable morphism and its kernel is just the set $X(\mathbb{F}_q)$ of points of X with coordinates in $\mathbb{F}_q$.

(c) Using (Ex. 4.7), show that $F' + \hat{F}' = a_X$ for some integer a, and that $N = q - a + 1$, where $N = \#X(\mathbb{F}_q)$.

(d) Use the fact that $\deg(m + nF') > 0$ for all $m, n \in \mathbb{Z}$ to show that $|a| \leqslant 2\sqrt{q}$. This is Hasse's proof of the analogue of the Riemann hypothesis for elliptic curves.

(e) Now assume $q = p$, and show that the Hasse invariant of X is 0 if and only if $a \equiv 0 \mod p$. Conclude that for $p \geqslant 5$ that X has Hasse invariant 0 if and only if $N = p + 1$.

Proof.

(a) Suppose X is locally isomorphic to $\operatorname{Spec} A$, where $A = k[x,y]/(f)$ and $f(x,y)$ is some degree three polynomial with coefficients in $\mathbb{F}_q \subset k$. Then X_p is locally isomorphic to $\operatorname{Spec}(k \otimes_q A)$, where the copy of k here has the qth-power Frobenius for the structure morphism as a k-algebra. The image of f under the identification

$$\begin{aligned} k \otimes_q k[x,y] &\cong k[x,y] \\ (b \cdot 1)x &\mapsto b^q x, \end{aligned}$$

gives a defining equation $f^{(q)}$ for X_q. Notice that $f^{(q)}$ is obtained from f by raising each coefficient of f to the qth-power. Since each coefficient of f is in $\mathbb{F}_q$, we see that $f^{(q)}$ and f are actually equal in $k[x,y]$. Hence, $X_q \cong X$ as schemes over k.
Under this indentification, $F' : X_q \to X$ is obtained from the restriction of the Frobenius on $\mathbb{P}^2$. The k-linear Frobenius morphism on an affine coordinate ring of $\mathbb{P}^2$ can be written as

$$\begin{aligned} F^* : k[x,y] &\to k[x,y] \\ g(x,y) &\mapsto g(x^q, y^q), \end{aligned}$$

which is exactly the ring homomorphism that gives the qth-power map on the coordinates.

(b) If $t \in \mathcal{O}_{P,X}$ is a local parameter of X, then $(1_X - F')^* t = t - t^p$. Since $d(t - t^p) = dt$, the map $1_X - F'$ is ramified nowhere. Hence, $1_X - F'$ is a separable morphism. Under the embedding $X \subset \mathbb{P}^2$, the kernel of $1_X - F'$ consists of the points fixed by the map on X obtained by the qth-power map on the coordinates of points of X. Since $\mathbb{F}_q$ is precisely the fixed field of k under the qth-power Frobenius, the kernel of $1_X - F'$ consists of the points of $X \subset \mathbb{P}^2$ with coordinates in $\mathbb{F}_q$.

(c) Recall from (Ex. 4.7), (Ex. 4.15), and (Ex. 4.16b) that

$$(1_X - F') \circ (\widehat{1_X - F'}) = N_X, \quad F' \circ \hat{F}' = q_X.$$

Expanding the left-hand side of the first identity gives

$$(1_X - F') \circ (\widehat{1_X - F'}) = 1_X - (F' + \hat{F}') + q_X.$$

Hence, we have

$$F' + \hat{F}' = q_X + 1_X - N_X.$$

(d) For all $m, n \in \mathbb{Z}$, we have the inequality

$$m^2 + n^2 q - mna = \deg(m - nF') > 0.$$

Rearranging gives

$$|a| < \left| \frac{m}{n} + \frac{n}{m} q \right|$$

The real-valued function $f(t) = t + q/t$ has a local extremas at $(\sqrt{q}, \pm 2\sqrt{q})$. The rational numbers are dense in the real numbers, so we conclude that $|a| \leqslant 2\sqrt{q}$. I wonder if it is every the case $a = 2\sqrt{q}$ when $\sqrt{q}$ is a perfect square.

(e) By (Ex. 4.15), the Hasse invariant of X is 0 if and only if $\hat{F}'$ is inseparable. If $t \in \mathcal{O}_{P,X}$ is a local parameter, then

$$(F' + \hat{F}')^* t = t^p + \hat{F}'^* t,$$

so $\hat{F}'$ is inseparable if and only if $F' + \hat{F}'$ is inseparable. The conclusion follows from the fact that a_X is inseparable if and only if $a \equiv 0 \mod p$. For $p \geqslant 5$, $a \equiv 0 \mod p$ and $|a| \leqslant 2\sqrt{p}$ can be both satisfied if and only if $a = 0$. What about the cases for $p < 5$?

□

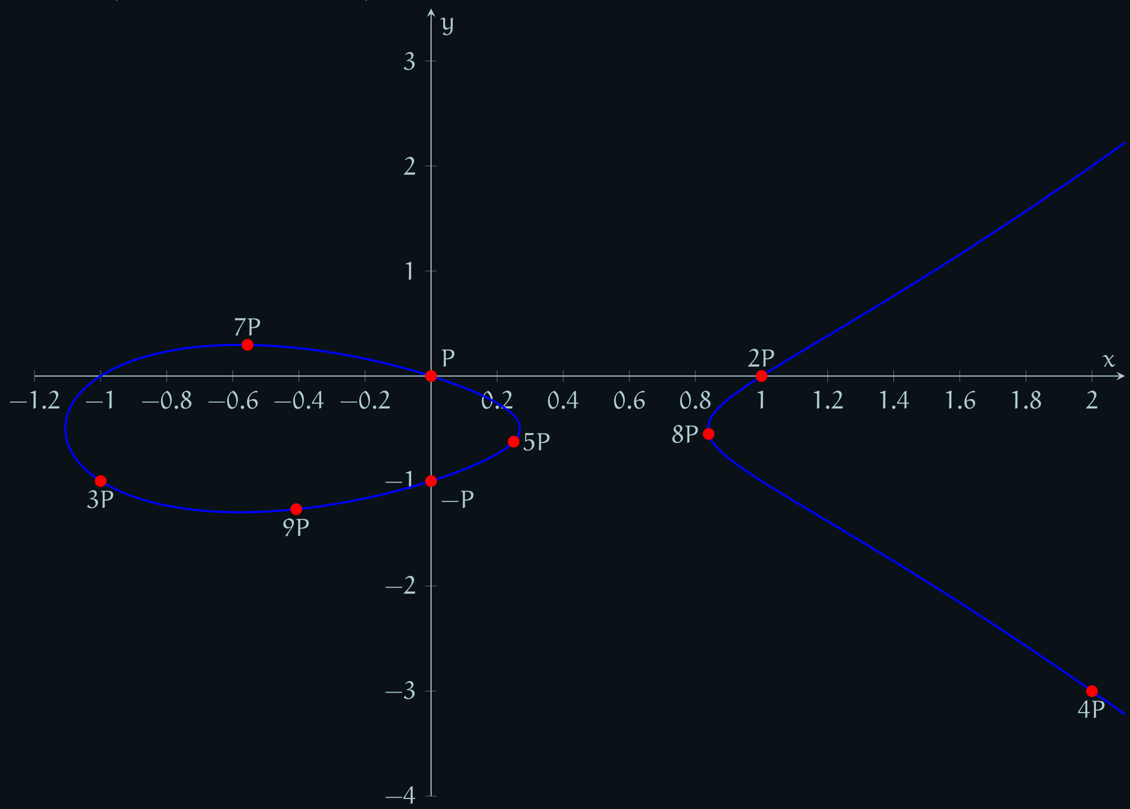
17. Let X be the curve $y^2 + y = x^3 - x$ of (4.23.8).

(a) If $Q = (a, b)$ is a point on the curve, compute the coordinates of the point $P + Q$, where $P = (0, 0)$, as a function of a, b. Use this formula to find the coordinates nP, $n = 1, 2, \ldots, 10$.

(b) This equation defines a nonsingular curve over $\mathbb{F}_q$ for all $p \neq 37$.

Proof.

(a) $P + Q = \left(-\dfrac{a+b}{a^2}, \dfrac{b(a+b)}{a^3} - 1 \right), \quad -P = (0, -1), \quad 2P = (1, 0)$



(b) Let X be a cubic curve in $\mathbb{P}^2$ given by an equation of the form

$$y^2 + a_1 xy + a_3 y = x^3 + a_2 x^2 + a_4 x + a_6, \quad a_i \in k.$$

Define the following quantities

$$\begin{aligned} b_2 &= a_1^2 + 4a_2 \\ b_4 &= 2a_4 + a_1 a_3 \\ b_6 &= a_3^2 + 4a_6 \\ b_8 &= a_1^2 a_6 + 4a_2 a_6 - a_1 a_3 a_4 + a_2 a_3^2 - a_4^2. \end{aligned}$$

The discriminant of X is given by

$$\Delta = -b_2^2 b_8 - 8b_4^3 - 27b_6^2 + 9b_2 b_4 b_6,$$

and X defines a nonsingular curve over $\mathbb{F}_q$ if and only if $\Delta \neq 0$ in $\mathbb{F}_q$. For our elliptic curve, the quantities b_i are

$$b_2 = 0, \quad b_4 = -2, \quad b_6 = 1, \quad b_8 = -1,$$

so the determinant of X is $\Delta = 37$. Hence, X defines a nonsingular curve over $\mathbb{F}_q$ for all $p \neq 37$.

$\square$

20. Let X be an elliptic curve over a field k of characteristic $p > 0$, and let $R = \mathrm{End}(X, P_0)$ be its ring of endormorphisms.

(a) Let X_p be the curve over k defined by changing the k-structure of X (2.4.1). Show that $j(X_p) = j(X)^{1/p}$. Thus, $X \cong X_p$ over k if and only if $j \in \mathbb{F}_p$.

(b) Show that p_X in R factors into a product of $\pi\hat{\pi}$ of two elements of degree p if and only if $X \cong X_p$. In this case, the Hasse invariant of X is 0 if and only if π and $\hat{\pi}$ are associates in R.

(c) If $\mathrm{Hasse}(X) = 0$ show in any case $j \in \mathbb{F}_{p^2}$.

(d) For any $f \in R$, there is an induced map $f^* : H^1(\mathcal{O}_X) \to H^1(\mathcal{O}_X)$. This must be multiplication by an element $\lambda_f \in k$. So we obtain a ring homomorphism $\varphi : R \to k$ by sending f to λ_f. Show that any $f \in R$ commutes with the (nonlinear) Frobenius morphism $F : X \to X$, and conclude that if $\mathrm{Hasse}(X) \neq 0$, then the image of φ is $\mathbb{F}_p$. Therefore, R contains a prime ideal $\mathfrak{p}$ with $R/\mathfrak{p} \cong \mathbb{F}_p$.

Proof.

(a) Some of the statements proved in (Ex. 4.16) hold here as well. In particular, under the idenficiation of the k-linear Frobenius morphism $F^* : k[x, y] \to k[x, y]$ defined by $x, y \mapsto x^p, y^q$, let $f(x, y)$ be a cubic polynomal that locally defines X in $\mathbb{P}^2$. The image of $f(x, y)$ under F^* is $f(x^p, x^p) = 0$, so if we write

$$f(x, y) = y^2 - x(x-1)(x-\lambda),$$

then

$$F^* f(x, y) = y^{2p} - x^p(x^p - 1)(x^p - \lambda).$$

Since k is algebraically closed, there exists a unique pth root $\lambda^{1/p}$ of λ. Thus, we can factorize

$$y^{2p} - x^p(x^p - 1)(x^p - \lambda) = (y^2 - x(x-1)(x - \lambda^{1/p}))^p.$$

The radical of the right-hand side gives the defining equation of X_p. Plugging in $\lambda^{1/p}$ gives $j(X_p) = j(X)^{1/p}$.

(b) If X admits a degree p endomorphism $\pi : X \to X$, then $X \cong X_p$ by (2.5). Conversely, if $X_p \cong X$, then p_X factors as $F' \circ \hat{F}'$, where $F' : X_p \to X$ is the k-linear Frobenius morphism (Ex. 4.7). By (Ex. 4.15), the Hasse invariant of X is 0 if and only if $\hat{F}'$ is an inseparable morphism. The Frobenius is the unique purely inseparable morphism of degree p (2.5), so if $X_p \cong X$, then F' and $\hat{F}'$ must differ by an automorphism of X, i.e., they are associates in R. If F' and $\hat{F}'$ are associates, then $\hat{F}'$ is an inseparable morphism since F' is inseparable. Hence, the Hasse invariant of X is 0.

(c) If $\mathrm{Hasse}(X) = 0$, then $\hat{F}' : X \to X_p$ is a purely inseparable morphism of degree p. By (2.5), X and the pth Frobenius twist $(X_p)_p \cong X_{p^2}$ of X_p are isomorphic as abstract k-schemes. Hence, $j(X) = j(X)^{1/p^2}$ by (Ex. 4.20a).

(d) Clearly f and F commute on the underlying topological spaces. The statement holds for any homomorphism between k-algebras, so it is immediate that f and F commute at the level of sheaves as well. If $\mathrm{Hasse}(X) \neq 0$, then the p-linear map of cohomology $F^* : H^1(X, \mathcal{O}_X) \to H^1(X, \mathcal{O}_X)$ is non-zero, and commutes with $f^* : H^1(X, \mathcal{O}_X) \to H^1(X, \mathcal{O}_X)$. In particular, $F^*(1) \neq 0$, and $(f \circ F)^*(1) = (F \circ f)^*(1)$. Expanding both sides gives

$$\lambda_f^p F^*(1) = \lambda_f \cdot F^*(1).$$

Since $F^*(1)$ is non-zero, we have $\lambda_f^p = \lambda_f$. Hence, λ_f is in $\mathbb{F}_p$.

$\square$